GNG2101

Design Project Progress Update

"CTRL Sync"
GROUP #3.1

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List of Acronyms and Glossary

Table 1. Acronyms

Acronym	Definition
ABS	Acrylonitrile Butadiene Styrene
DFX	Design for X
FPS	First Person Shooter
LCA	Life Cycle Analysis
PLA	Polylactic Acid

Table 2. Glossary

Term	Acronym	Definition
Add-on Attachment		A device that attaches to an existing product to add functionality.
Adaptive Device		A device designed to assist users with physical impairments.
Cradle-to-Grave		An LCA approach covering a product's entire life cycle from creation to disposal.
Design for X	DFX	A design approach that optimizes a product for specific objectives.
First Person Shooter	FPS	A game genre played from the character's point of view that involves aiming and shooting.
Life Cycle Analysis	LCA	A method for evaluating environmental impacts across a product's life cycle.
Modular Design		A design using replaceable components to improve repairability.

1 Introduction

This deliverable contains the foundation for the prototype phase, up until our conclusion via presentation at design day. The contents of the file span a first iterative prototype, in which we relate the DFX in previous submissions to our current one, expanding on the adjustments needed, as well as what obstacles came up that caused us to iterate our initial concept. We expand on this in our second prototype and then being to consider the economic ramifications of our project. We conclude with a user manual to elaborate instruction for our project.

2 Prototype 1, Project Progress Presentation, Peer Feedback and Team Dynamics

2.1 Prototype 1

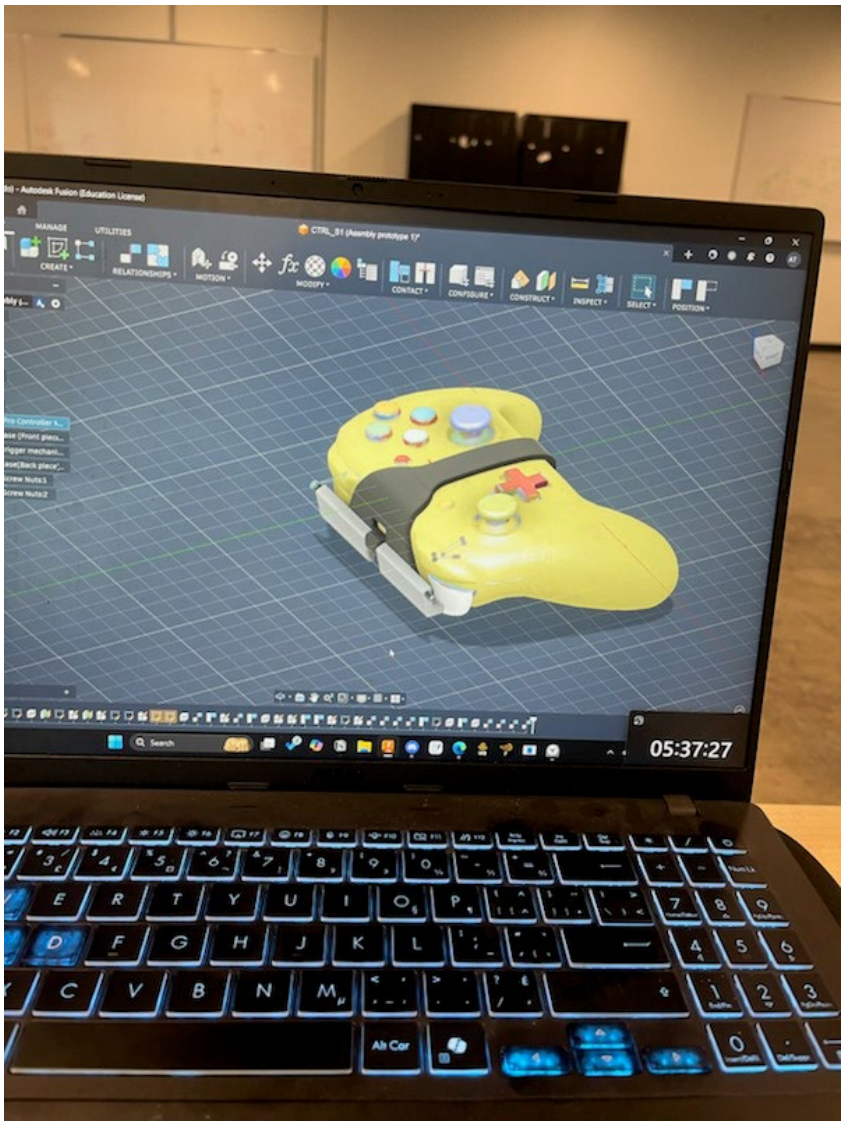

Figure 1: CAD Design Model of Our Prototype.

Figure 2: Rear View of Our Prototype.

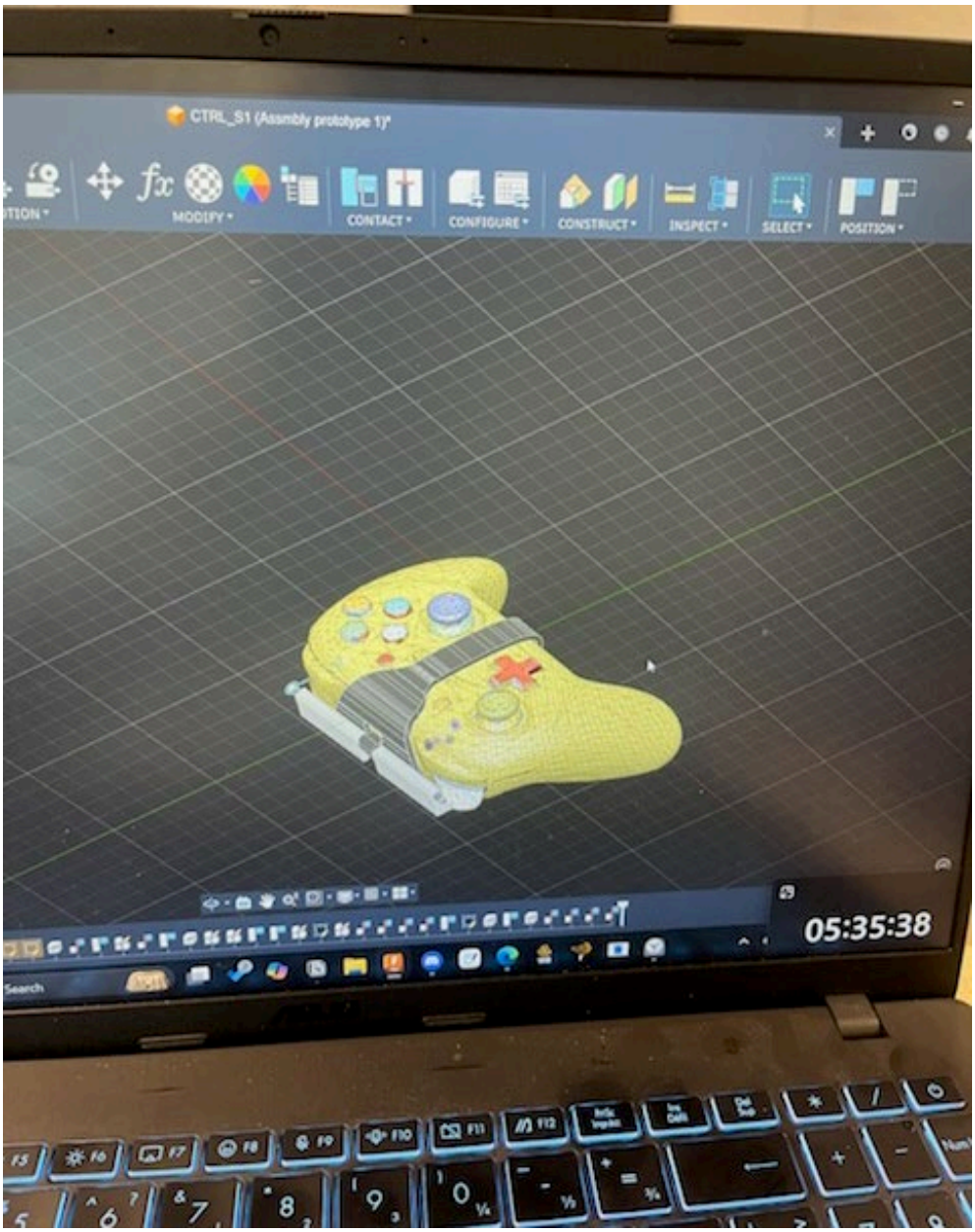

Figure 3: CAD Rendering of Our Prototype.

Figure 4: Our Prototype, alongside A Mold Controller.

We assume, like we did in previous deliverables, that the controller operates without the use of external intervention of any electronic components. This is consistent with DFX for accessibility, mentioned in deliverable B. However, adjustments needed to be made to the design that we had in mind. Originally, we had environed that the operator of the apparatus would use it in their lap, allowing external access to the triggers via a pushing motion. But this would reduce the presence of the actuation force, making it hard for consistent measurement unless at a specific angle. Our new approach allows an attachment on top of the device, allowing the user to set it on the table and use the device.

Table 3: Expected Testing Values

Test Category	Product assumption being verified	Intended Method	Target value
Trigger Actuation Force	Attachment reduces effort needed to press L2/R2 triggers	Add small known weights until trigger activates	≤ 5 N required force
Reach Extension Length	Attachment allows the client to reach triggers comfortably	Measure extension using calipers	30–45 mm extension
Installation & Setup	Device can be installed easily without tools	Time multiple simulated installs	< 90 s setup, tool-free
Comfort & Material Feedback	Material is comfortable for gameplay	User held different printed samples and rated comfort (1–5)	≥ 4/5 comfort rating

2.2 Project Progress Presentation

Our presentation can be found here. It's included separately in this as well.

2.3 Project plan update

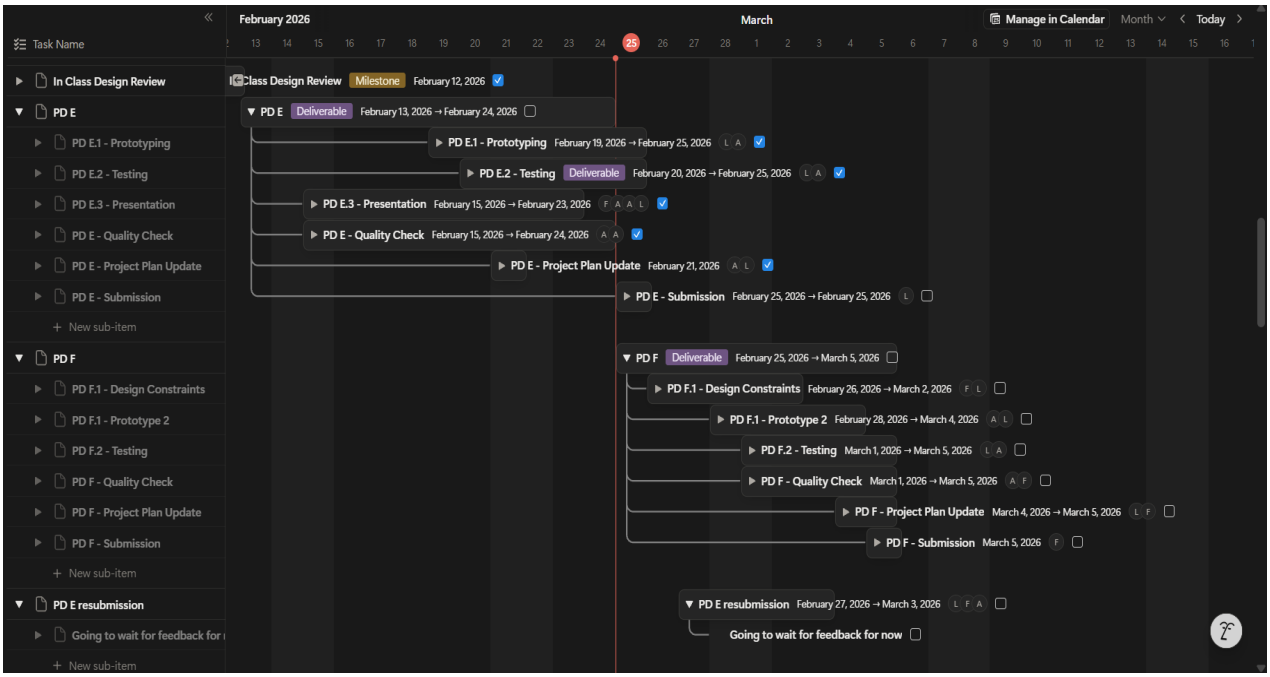

Figure 5: Updated Gantt Chart (as of Feb 25,2026)

2.4 AI Appendix

Given the nature of this deliverable, no AI was used for this portion of the assignment.

3 Design Constraints and Prototype 2

3.1 Design constraints

The first important design constraint that is considered in this project is Design for Durability because the attachment must be capable of being used more than once during the game and performing the required functions without collapsing or causing the worsening of its quality. As it is in direct contact with controller triggers via mechanical lever arms, it is under high frequency cyclic loading which may cause wear or structural failure of the material due to repeated loading. This was mitigated by designing it with strong materials, such as 3D-printed plastics (e.g., PLA or ABS) that have enough thickness and reinforcement of the high-stress zones, especially around the lever arms and areas of connection. Moreover, the design geometry was modified with the idea of decreasing stress concentrations by not having sharp corners and rather smoother redistribution of loads throughout the construction. In the prototype stage, wear and stability observations aided in the improvements to be made including reinforcement of the attachment points and making sure the fit was secure because it would be lost easily otherwise. To prove that the design can withstand durability we took into consideration the projected force to be used in the process of actuation by the trigger (5 N) and made sure that the material and the structure could safely absorb recurring load without deformation and breakage. We also dropped them from a certain height to see if it would break or expose areas that are more likely prone to break. Although complete fatigue testing over the long run was not performed, repeated manual testing and examination indicated that there was only one sign of failure which was at the bottom of the attachment. Even though there was one failure, most of it was still in tack and indicated that

the design was reliable. Right now, we are thinking of a way to fix one failure, and we are positive that we will be able to fix it by Design Day.

The second design constraint is Design for Assembly that deals with the ease of installation and removal of the attachment without having to involve much effort and no equipment. This is particularly necessary because the client and the target users might lack dexterity or strength. The first prototype had an attachment system which made use of a clip-in system that was based on geometric position and friction to ensure that the attachment was held in place on the controller. Nonetheless, this design had some problems during testing and usage, namely it was not easy to fit correctly, used more force than anticipated and in other instances difficult to take off without harming or creating uneven forces. This lowered its overall usability and rendered the device less convenient as independent usage. This led the design to be enhanced by the replacement of the clipping system with a magnetic attachment system whereby the device would easily fit and be detached with very little effort. This modification aids the more modular design and substantially enhances ease of use. This constraint is fulfilling as shown in the target requirements, which include no equipment needed to install and less than 90 seconds was taken to install the tool. To check this, the actual client tested it in which he was required to install and uninstall the device on a few instances. According to this test, the client indicated that the installation procedure was not complicated and was simple to learn, and that the process of removing the device would not need much effort. It was also recorded that the client had installed it in less than 50 seconds on average, which is a good indication of validating our design change.

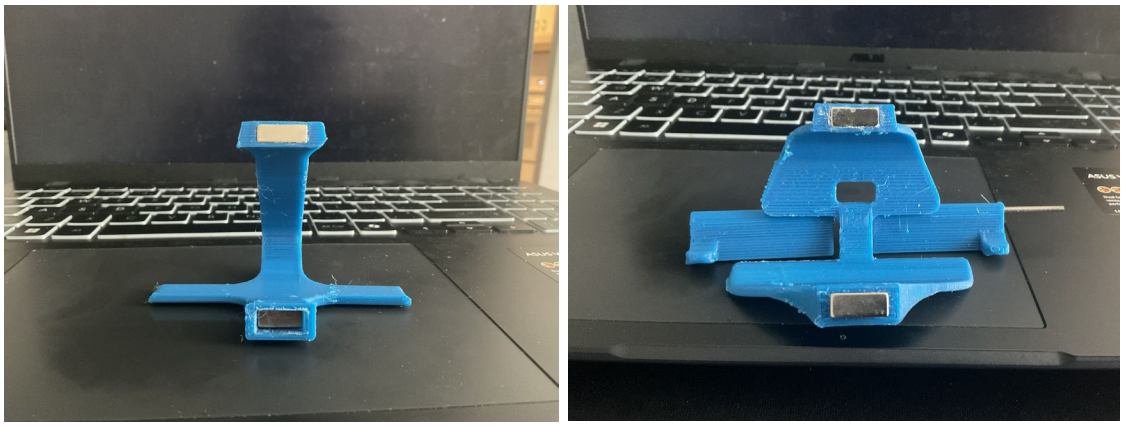

Figure 6: Improved Prototype Showing Magnetic Attachment System for Easy Assembly

To conclude, the design was effective in complying with the requirement of both durability and assembly by being improved and tested repeatedly. The device demonstrates sufficient strength to withstand repeated use while remaining easy for the user to install and remove independently. These results justify the fact that the second prototype design is dependable and applicable in the real world.

3.2 Prototype 2

Client Feedback and Design Changes

Based on feedback received during the second client meeting, two major improvements were identified for the next iteration of the design. First, the client suggested that we should design one of the paddles on the attachment (specifically the left paddle) to have an auto-lock feature rather than the traditional holding feature. The client had also stated that we should keep the right paddle as it is. This is important as the game they love to play does not have a feature in

the settings to have this mechanism. Also, this would reduce strain and improve comfort during extended gameplay. Second, the client mentioned that both material options tested were comfortable and therefore did not have a strong preference between them. As a result, the responsibility now lies with the team to select the most suitable material for long-term durability and repeated use. Though both feedback would give good guidance towards further enhancements, the client also said that they would be happy with the present design as it is as these are just extra things we could do with whatever time we have left before Design Day. Overall, these lessons contributed to making further improvements and ensuring that the design is already efficient and popular.

Critical Product Assumptions

An important assumption that is yet to be totally tested is if the magnetic attachment system will retain their magnetic strength over time, meaning they will continue to provide a strong and reliable connection even after long-term use. This is directly connected to Design for Durability and Design for Reliability, as the magnets should never let the attachment fall and not work anymore. Nevertheless, it will be challenging to test the long-term lifespan of the magnets due to the time limit of the current project, and it is believed it will serve a quite decent duration under the normal usage conditions.

Prototype Description and Functionality

The most recent prototype includes the one-handed controller attachment which can enable two triggers to be used with one hand by providing an extended paddle system. The controller works by sending the input of the available paddles to the triggers of the controller through the use of mechanical lever arms, which saves the necessity of the use of both hands. The final prototype has a magnetic attachment system installed, making it easier to install and remove quickly, and more accessible and easier to use than the previous design, the clip-in. The structure becomes stronger at the high-stress areas, and the forms are smooth to create less stress concentration and enhance the duration. Also, the design allows the paddles to be placed in an ergonomic manner that makes them easily accessible and allows them to be used numerous times. Additionally, the layout ensures that the paddles are ergonomically positioned for comfortable reach and repeated use, requiring less finger extension to activate the paddles. This reduced reach helps minimize strain on the user's fingers during gameplay, especially over longer periods. In general, the prototype aims at the higher accessibility of gaming, preserving functionality, comfort, and reliability.

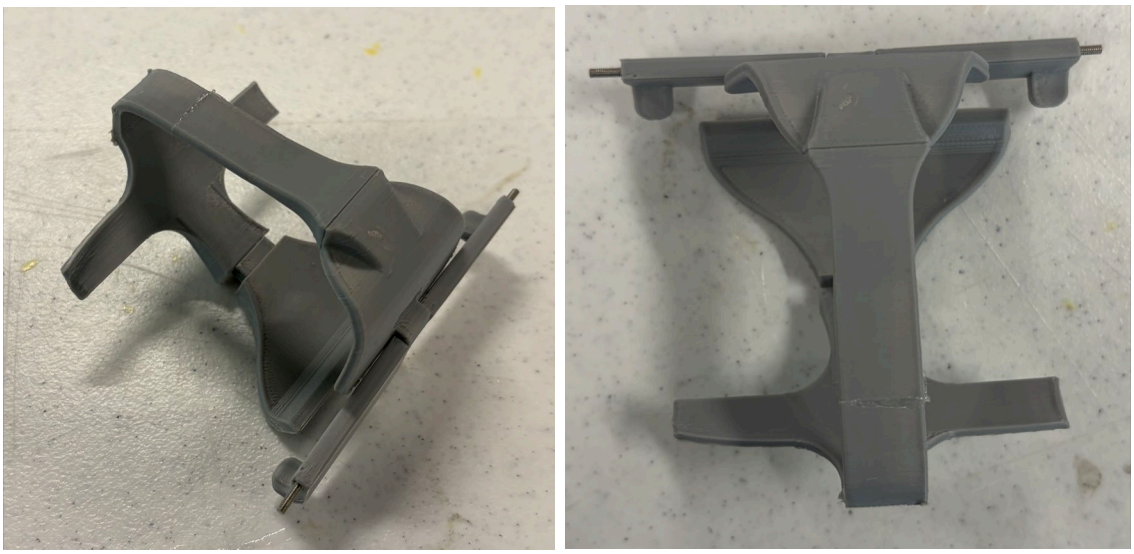

Figure 7: Final Prototype of Controller Attachment

Prototype Testing and Performance Evaluation

Table 4: Prototype Performance Evaluation Against Target Specifications

#	Metric Test	Unit	Expected Results	Actual Results
1.	Maximum trigger actuation force through the attachment.	N (Newtons)	5.5 ± 0.5	≈ 4.8
2.	Effective trigger reach extension.	mm	30 - 45mm	≈ 40.3
3.	Material comfort and surface texture rating.	1–5 scale	≥ 3.75/5	≥ 4/5 (Both for PLA and ABS)
4.	Installation time	sec	≤ 90 sec	≈ 45 (On average with 4 different attempts)
5.	Tools required	Y/N	No	No

The table presents the results of the comparison of the approximations regarding the target specifications with the real measurements of the prototype. All in all, the prototype satisfied or surpassed the majority of design requirements. The force to actuate the trigger was marginally less than that of expectation thus requiring less effort on the part of the user and this enhances usability. The extension of the reach was found in a range desired, and it was comfortable to access the paddles and minimized any strain on the fingers. Both PLA and ABS met the target of material comfort rating, indicating that the two materials could be used in the interaction of a user. Also, the magnetic attachment system proved productive as the time spent on installing the device was much less than the maximum requirement and did not need any tools. These findings indicate that prototype works in a very reliable fashion and it is properly aligned with the targeted design aims, but there are some areas that can be refined.

Client Presentation and Feedback Plan

For the next client meeting, the team plans to present the updated prototype and demonstrate its key features, including improved durability and overall ease of use. The team will also introduce the proposed auto-lock feature for the left paddle to gather the client's opinion on its usefulness and comfort. During the session, feedback will be collected on ergonomics, particularly finger strain and reach, as well as overall usability during extended use. Additionally, we will get the client to try and damage the product to showcase its improved durability. We will also finally get him to play his favourite game with the attachment. This feedback will be used to guide final design refinements and ensure the product meets the client's needs before Design Day.

3.3 Project plan update

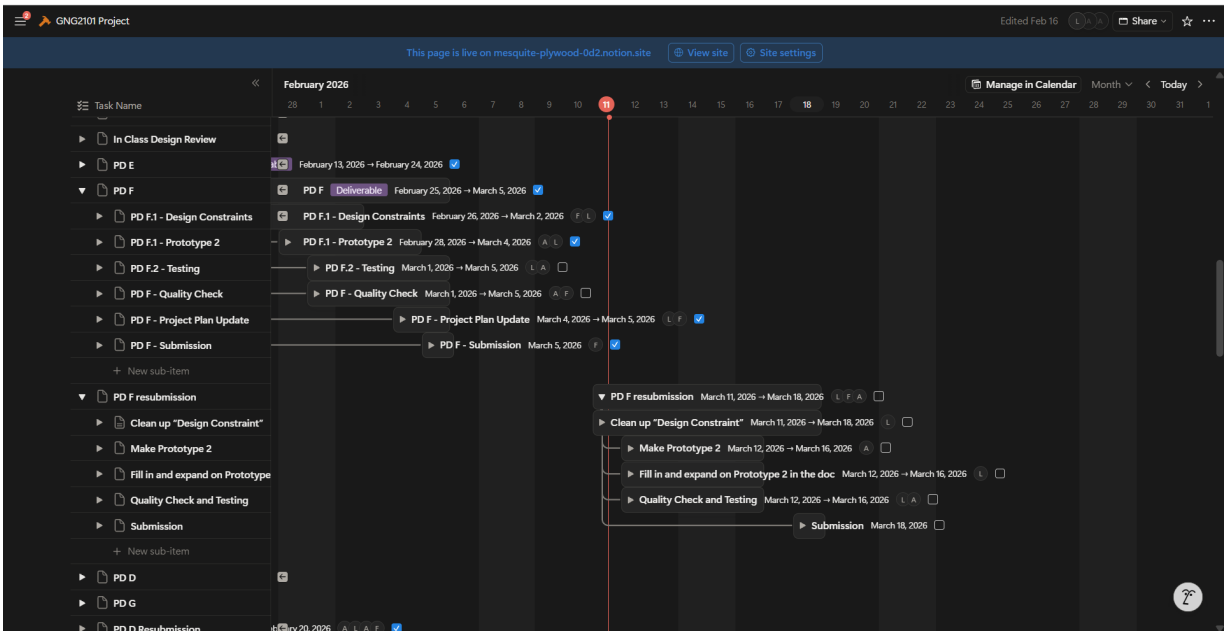

Figure 8: Updated Gantt Chart (as of March 11, 2026)

3.4 AI Appendix

AI was used in our project deliverable in a limited way, mainly to help us better understand certain topics and to fix small grammar or wording issues in the report. It was also helpful for getting a clearer idea of how different sections of the deliverable fit together and what was expected in each part. In some cases, AI was used to help make our writing more condensed while keeping the information clear and concise. Additionally, it assisted in the creation of titles, tables and section headings to ensure consistency.

It is also worth mentioning that the key ideas or the main body of our work were not created with the help of AI. Our team devised all design and testing choices as well as analysis by their own knowledge, conferences, and practical experience with the prototype. It was implemented as an aid tool to enhance clarity and organization, but the final content contains what was first thought of by our team and the work that went into it.

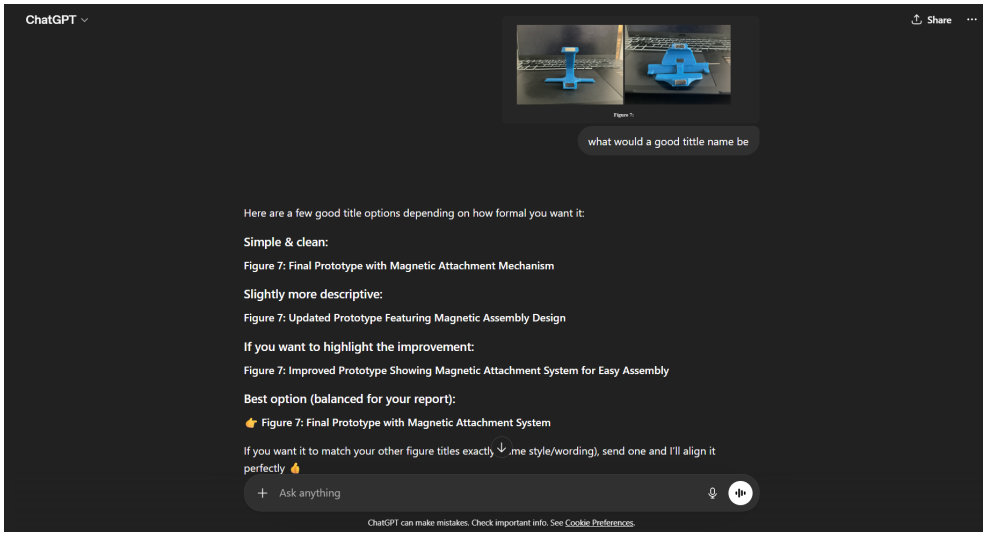

Figure 9: Example of AI-Assisted Figure Title Generation

4 Economic and IP Considerations

4.1 Economics report

4.2 Intellectual property report

4.3 Project plan update

Add a screenshot of your gantt chart.

4.4 AI Appendix

5 Design Day Pitch and Final Prototype Evaluation

Write your design day pitch and plan your prototype demo. Add a link to your slides if you have some.

6 Video and User Manual

6.1 Video pitch

Add link to video.

6.2 User manual

See separate template for the user manual. Do not write the content here.

6.3 AI Appendix

7 Conclusions

Summarize your lessons learned and your work related to your project. Discuss any outstanding issues or implications for the project.

8 Bibliography

Insert your list of references here.